GFAz: Order-of-Magnitude Pangenome Analytics by *Computing over Compression*

Taolue Yang Youyuan Liu Bo Jiang Xinghua Shi Sian Jin
Department of Computer & Information Sciences, Temple University

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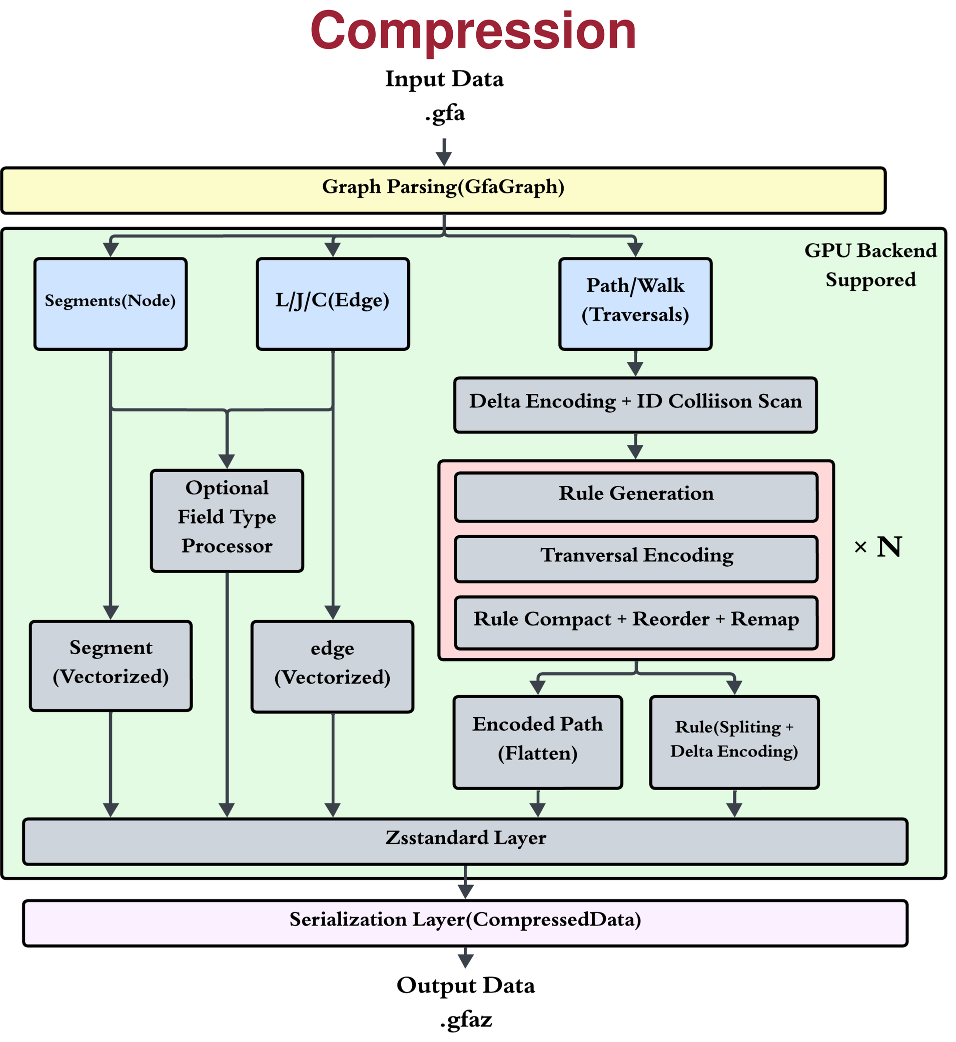
84×	5.3 GB/s	369×	99.99%
smaller	decompression	faster analytics	VCF concordance
369 GB → 4.5 GB	CPU, 32 threads	Panacus 245 min → 40 s	vs. `vg deconstruct`

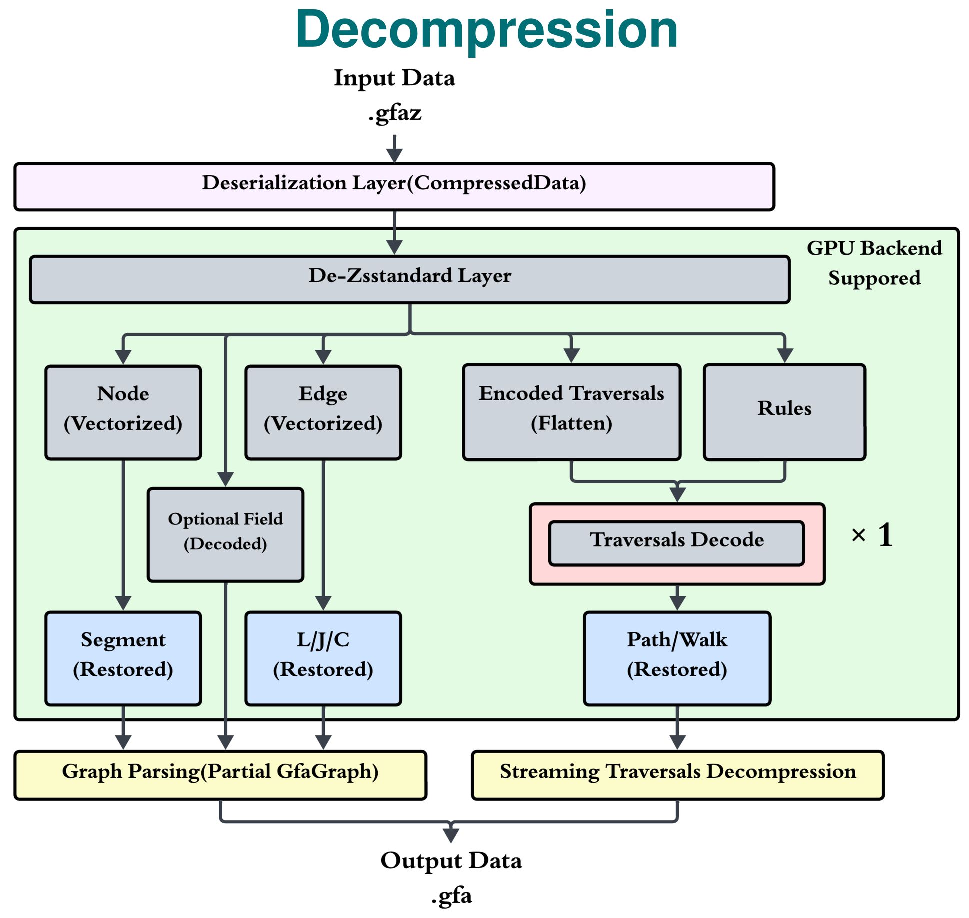
1 Compression engine

lossless · full GFA semantics · CPU + GPU

Compression



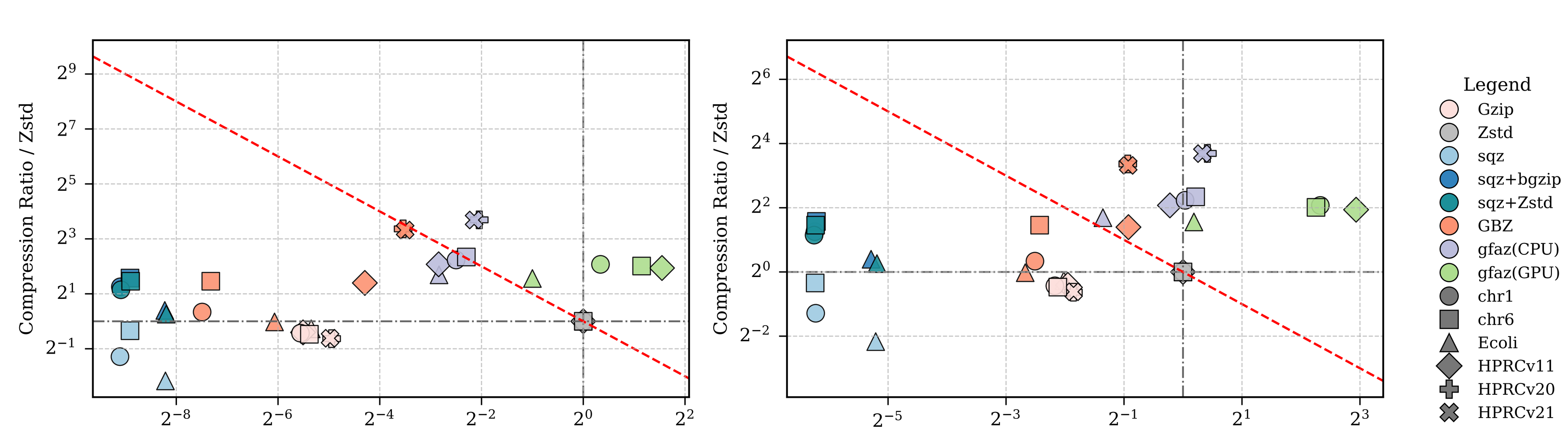
Decompression



Traversals (>90% of a GFA) collapse under **iterative 2-mer grammar compression**; every other record type — optional fields included — gets **columnar, type-specific** encoding. Both directions are linear-time and fully parallel.

Dataset		Gzip	Zstd	sqz	sqz+bgz	GBZ	GFAz CPU	GFAz GPU
chr1	Ratio	5.59	7.54	3.09	18.0	9.52	**35.4**	31.7
	Co.	46.2	2178	3.95	3.97	12.1	1320	**2754**
	De.	359	1618	21.6	21.4	284	2307	**8124**
chr6	Ratio	5.04	6.99	5.51	20.8	19.2	**35.4**	28.2
	Co.	41.0	1712	3.56	3.56	10.7	1355	**3791**
	De.	348	1515	20.1	20.4	281	2943	**7230**
E. coli	Ratio	4.69	5.67	1.26	7.46	5.58	**18.4**	16.7
	Co.	33.3	**1356**	4.57	4.53	20.2	226	678
	De.	310	1258	34.0	32.2	197	834	**1430**
HPRC v1.1	Ratio	4.02	5.32	–	–	14.0	**22.4**	20.4
	Co.	36.4	1657	–	–	84.5	291	**4843**
	De.	319	1234	–	–	650	2292	**9435**
HPRC v2.0	Ratio	4.19	6.49	–	–	66.8	**83.8**	76.4
	Co.	49.1	**1514**	–	–	130	555	–
	De.	342	1240	–	–	648	**5426**	–
HPRC v2.1	Ratio	4.19	6.43	–	–	64.2	**82.8**	74.2
	Co.	48.9	**1540**	–	–	136	538	–
	De.	343	1241	–	–	652	**5325**	–

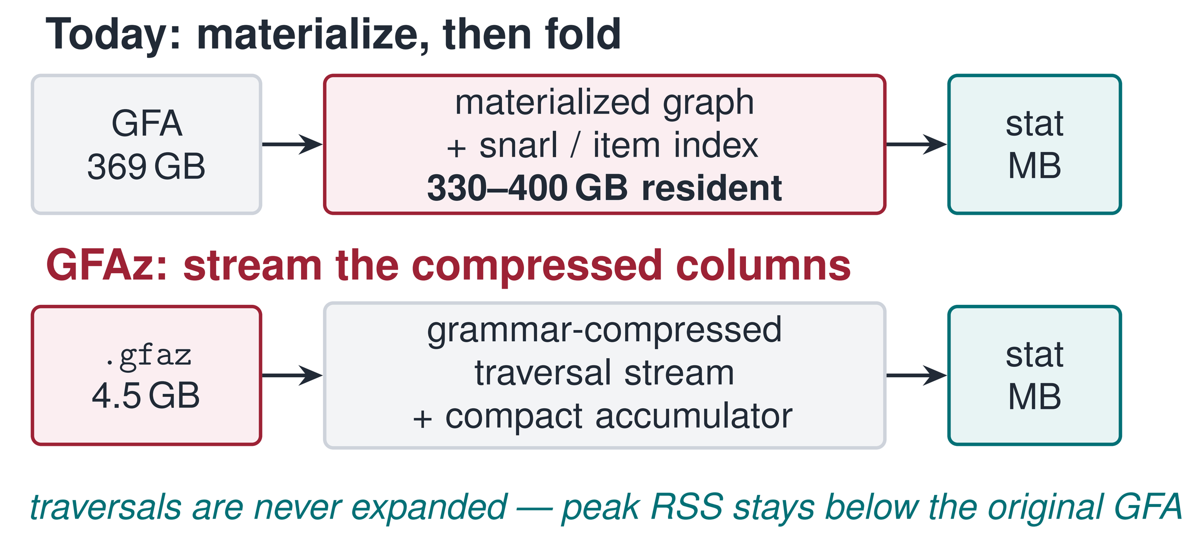
Compression ratio, and compression (**Co.**) / decompression (**De.**) throughput in MiB/s. GFAz CPU at 32 threads; CPU compression is end-to-end CLI wall time. **Best ratio on every dataset**, and the fastest decompression wherever the GPU backend fits in device memory. Threadripper PRO 9955WX, RTX Pro 6000, 512 GB DDR5.



Ratio–speed Pareto front, normalized to Zstd = 1.0. The dashed diagonal is iso-efficiency: **up and to the right beats the baseline**. GFAz sits above and right of every alternative on decompression, and the GPU backend dominates outright.

2 Compute engine

run the analysis on the compressed container



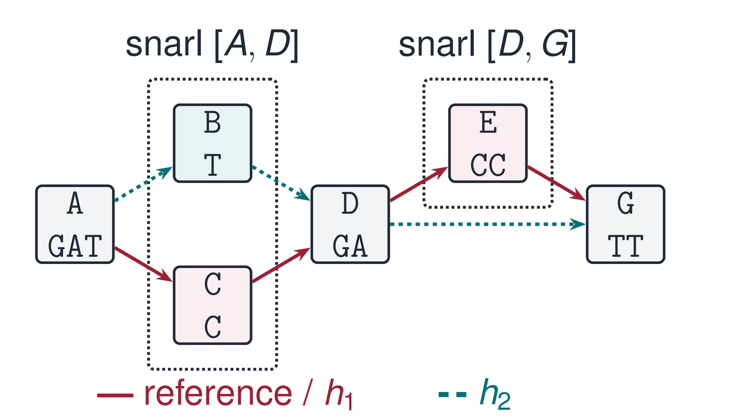

```
StreamDecodedNodes(HapSlice, Visit)
```

One primitive backs every analysis. Rule lookup is an **array index**, not a hash; a negative rule ID means **reverse complement**, so one rulebook serves both strands.

GFAz command	Reproduces	Passes	Accumulator
`deconstruct`	`vg deconstruct`	1 (+topo)	per-site allele table
`growth`	Panacus	1	coverage → histogram
`depth`	`odgi depth`	1	per-node counters
`stats`	`odgi stats`	0	metadata totals
`pav`	`odgi pav`	3	node→groups CSR
`similarity`	`odgi similarity`	2 (+join)	node→groups CSR

Six analyses, one decoder — they differ only in the accumulator. A new analysis is a new visitor.

`deconstruct`: GFA → VCF with no graph



#CHROM	POS	REF	ALT	INFO	h1	h2
chr1	4	C	T	AC=1;AN=2;AF=0.5	0	1
chr1	6	ACC	A	AC=1;AN=2;AF=0.5	0	1

Phase 1 recovers snarls from *topology alone* (biconnected decomposition, $O(V{+}E)$, decodes zero haplotypes). **Phase 2** streams each haplotype once through a boundary state machine; a reverse boundary pair normalizes inversions. No graph, no general snarl index.

Analysis	Graph	Baseline	GFAz	Speedup	Mem.
`deconstruct` *vs.* `vg`	chr1	805 s / 30.3 GB	11.3 s / 8.3 GB	**71**×	**3.7**×
	HGSVC3	132.9 min / 397 GB	8.7 min / 51.9 GB	**15**×	**7.6**×
`growth` *vs.* Panacus	chr1	18.3 s / 6.16 GB	0.74 s / 0.66 GB	**25**×	**9.3**×
	v2.0	245 min / 327 GB	39.8 s / 12.9 GB	**369**×	**25**×
`pav` *vs.* `odgi`	chr1	3499 s / 31.9 GB	11.7 s / 9.5 GB	**299**×	**3.4**×
	chr6	4759 s / 19.4 GB	7.8 s / 6.4 GB	**613**×	**3.0**×

Wall-clock / peak RSS at 16 threads. **Outputs match the baselines:** `growth` reproduces Panacus' curve exactly, `deconstruct` agrees with `vg` to within 0.2% of records, `pav` matrices are structurally identical to `odgi`'s. On the 358/369 GB HPRC graphs `vg` and `odgi` *cannot load the input at all* — GFAz runs from a 4.5 GB container.

A pangenome graph does not have to be unpacked to be queried.

One container gives **best-in-class compression**, **GB/s decompression**, *and* a **compute substrate** that runs six analyses one to three orders of magnitude faster than the standard tools.

Open: whole-genome `pav` still exceeds memory on the two largest HPRC graphs — its node-set index is not yet compressed.

Open source: `github.com/babyplutokurt/GFAz` · **Contact:** `taolue.yang@temple.edu`

Paper: Yang, Liu, Jiang, Shi, Jin. *GFAz: Order-of-Magnitude Pangenome Analytics by Computing over Compression.* ICS '26. `doi:10.1145/3797905.3807870`

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code

paper